$k = 5$, 2-torsion Edwards route, end to end with F4 trace replay

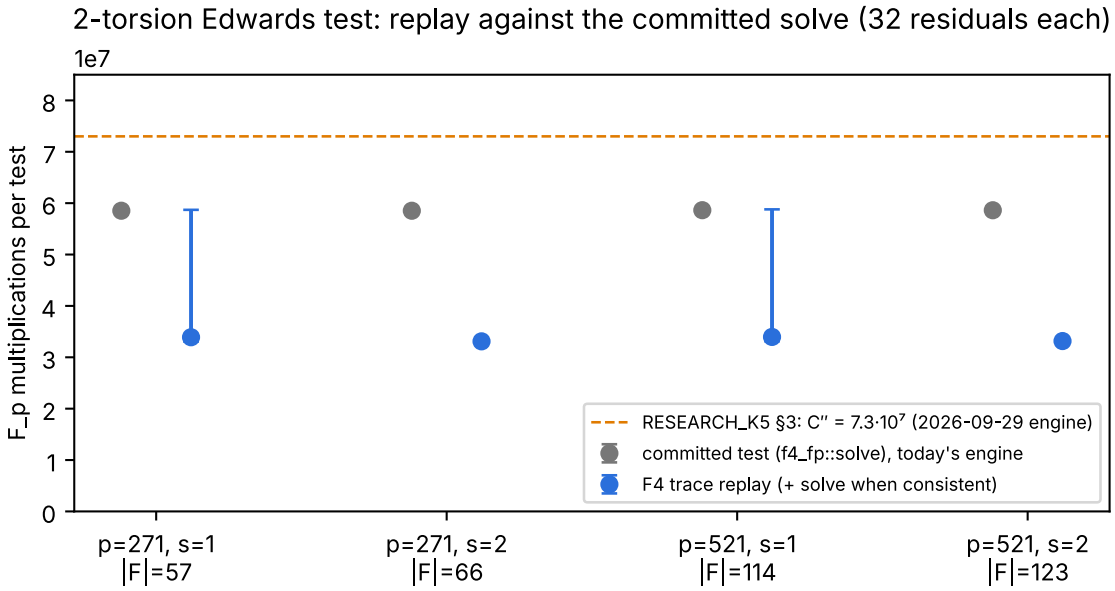
Provisional. Built from the records present when it was rendered: DLP arms without a final row are shown as incomplete. The protocol's remaining arms are still running.

2026-10-15 · `research/joux_vitse_n1_20261015/` · protocol `PROTOCOL_EDWARDS.md` (frozen at `3a8599b3f`), deviations `data/edwards/DEVIATION.md` · code `src/cryptanalysis/joux_vitse_edwards.rs`, `examples/joux_vitse_edwards.rs`, summary `examples/joux_vitse_edwards_summary.rs` · construction unchanged from `jv_quintic_edwards.rs` (RESEARCH_K5_TORSION_JOUX_VITSE.md). Checks failed so far: **0**.

The test: trace replay against the committed solve, same residuals

run	$\|F\|$	committed solve (mean)	replay (mean)	factor mean	min	max	replay div.	checks failed
cost_p271_s1.jsonl	57	$5.85{\cdot}10^7$	$3.39{\cdot}10^7$	1.74	1	1.77	0	0
cost_p271_s2.jsonl	66	$5.85{\cdot}10^7$	$3.31{\cdot}10^7$	1.77	1.77	1.77	0	0
cost_p521_s1.jsonl	114	$5.86{\cdot}10^7$	$3.4{\cdot}10^7$	1.74	1	1.77	0	0
cost_p521_s2.jsonl	123	$5.86{\cdot}10^7$	$3.32{\cdot}10^7$	1.77	1.76	1.77	0	0

On the committed curves ($p = 271, 521$, the bases of RESEARCH_K5 §3), the committed test now costs $5.85 \cdot 10^7$ (the note's $7.3 \cdot 10^7$ was measured on the 2026-09-29 engine) and the replay 3.3–$3.4 \cdot 10^7$: a factor **1.74–1.77**, below the 3–10 × the note priced. Every replayed residual's decompositions equal the committed solve's and the meet-in-the-middle oracle's.



Cost of one test per residual: committed solve and replay (mean, min–max), with the note's figure.

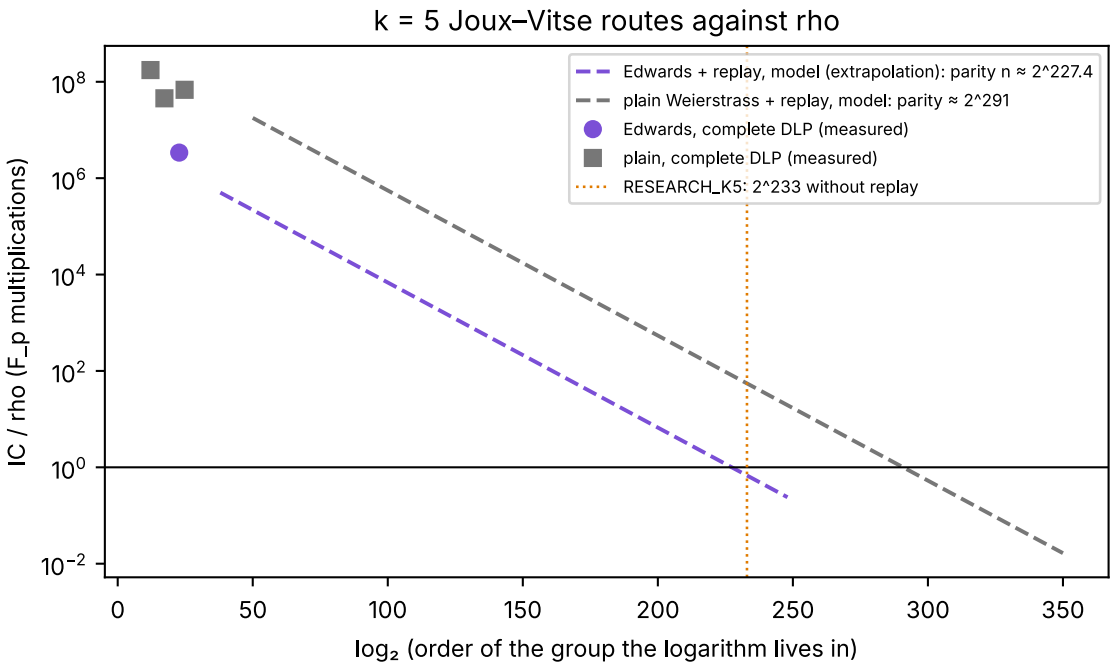
Complete DLPs, paired with rho on the same target

run	$\log_2 n$	$\|F\|$	trials	rel.	trials/rel.	÷ census	IC online	IC / rho	IC / exp. walk
dlp_p31_s1.jsonl	22.8	7	105344	8	13168	0.52 [0.26, 1.19]	$5.93{\cdot}10^{12}$	$3.39{\cdot}10^6$	$3.91{\cdot}10^6$
dlp_p41_s1.jsonl		6	running	1 so far	—	census 241333	—	—	—
dlp_p61_s1.jsonl		14	running	11 so far	—	census 26369	—	—	—

The census column is the measured trials per relation over the census expectation $n/\left(8\binom{|F|}{4}\right)$ (RESEARCH_K5 §3), with an exact Poisson 95 % interval on the relation count; it is an expectation, not a bound. At $p = 31$ the complete DLP recovered the planted logarithm (verified $kG = Q$; rho on the same target verified too) at **$3.39{\cdot}10^6$ × rho**; the plain Weierstrass route at the same p measured $6.8 \cdot 10^7$. About 3 % of residuals ($\approx 1/p$) give a consistent system; each pays the replay and the full solve, so this run's mean trial is $5.63{\cdot}10^7$ against the replay's $3.4 \cdot 10^7$. `dlp_p41_s1` drew a six-point base (census $\approx 2.4 \cdot 10^5$ trials per relation) and is expected to reach its 10-hour cap.

Where it would meet rho (extrapolation)

On the measured replay cost ($3.35{\cdot}10^7$), the census rate $1/(192p)$, $|F| + 1 \approx p/4$ relations and rho with the negation map at $c_{\text{add}} = 452$: parity at $n \approx 2^{227.4}$ ($p \approx 2^{45.9}$), against the note's 2^{233} without replay. At the measured $p = 31$ ratio to the census (0.52, interval containing 1) it would read $2^{217.8}$; that is a sensitivity, not the estimate. Either way the route stays 10^5–10^6 times rho at every size this repository can run.



IC / rho: complete DLPs (Edwards, circles; plain Weierstrass, squares) and the two models (dashed; extrapolation).